

SACHINI DILRANGI

Software Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with 1+ year of experience building full-stack applications with Java/Spring Boot, React.js, and Next.js — including its API routes and the Node.js tooling ecosystem — covering RESTful API and microservices design, SQL/NoSQL database work, and frontend state management. Also builds AI-integrated systems, from ensemble NLP-based plagiarism detection to a hybrid-search RAG pipeline built from scratch. Comfortable with Docker, CI/CD, and Git-based DevOps workflows, with a strong Test-Driven Development mindset and experience delivering in Scrum Agile environments. Final-year Software Engineering undergraduate with a track record of shipping tested, production-quality code.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL

Backend: Java, Spring Boot, Microservices Architecture, RESTful API Design, WebSockets, JWT Auth, Next.js API Routes, FastAPI

Databases: SQL (PostgreSQL, MySQL), MongoDB (Atlas), Redis, Firebase, Query Optimization

AI/ML Integration: LLM API integration (Groq), prompt engineering, NLP similarity algorithms (TF-IDF, Jaccard, N-gram), retrieval-augmented generation (hybrid BM25 + semantic search via Reciprocal Rank Fusion, built from scratch), local LLM inference (Ollama)

Frontend: React.js, Next.js 15, Redux, TanStack Query, Tailwind CSS, HTML5, CSS3

Cloud & DevOps: AWS (S3), Docker, GitHub Actions (CI/CD), Git, Vercel

Testing & Quality: TDD, Jest, React Testing Library, Unit/Integration Testing, Scrum/Agile, Jira, Figma

Currently expanding: vector databases (pgvector), broader AWS services (ECS, Lambda, RDS), Go, standalone Node.js/Express, and agentic AI tooling (Google ADK, n8n, Gemini Pro)

PROFESSIONAL EXPERIENCE

Associate Software Engineer | Cloud-eDesign *Aug 2025 – Apr 2026*

- Delivered frontend, backend, and admin dashboard features across 15+ client projects using Next.js and Spring Boot, with REST APIs and database-backed integrations throughout.
- Built RESTful APIs backed by relational and NoSQL databases; participated in code reviews and wrote unit/integration tests to maintain code quality.
- Used Git and CI/CD workflows within a Scrum Agile, remote-first team to ship clean, well-documented, production-ready code.

Software Engineer Intern | Cloud-eDesign *Feb 2025 – Aug 2025*

- Assisted in developing and testing web application features under senior engineers, gaining hands-on experience with the team's front-end and back-end stack.
- Contributed to bug fixes and troubleshooting, and learned Agile practices, Git workflows, and code review standards.

PROJECTS

Smart LMS — Versioned Submission & AI Feedback System | *Final-Year Research Project (SLIIT) — Spring Boot, PostgreSQL, Redis, Next.js, TypeScript, Microservices Aug 2025 – May 2026*

- Sole developer of a Spring Boot microservices module (submission management, AI feedback, plagiarism detection, version control) within a 4-person capstone system, following TDD practices with unit/integration tests throughout.
- Built RESTful APIs and an ensemble plagiarism-detection engine (TF-IDF, Jaccard, N-gram Dice) achieving a 91% true-positive rate at 8% false positives, with real-time WebSocket alerts.
- Integrated Groq's LLM API for real-time feedback (avg. under 5s latency) and built a consistency-enforcement engine that cut feedback contradictions from 23% to 6%; deployed with 99.2% uptime over an 8-week production run.

simple-RAG — Retrieval-Augmented Generation Pipeline / *Personal Project — Python, FastAPI, sentence-transformers, NumPy, Ollama, Next.js*

- Built a full RAG pipeline from scratch with no LangChain or LlamaIndex — custom chunking with configurable overlap, a hand-written NumPy vector store, and hybrid search combining BM25 with semantic embeddings via Reciprocal Rank Fusion (RRF).
- Added a relevance threshold that refuses to answer when nothing in the corpus is relevant, with local LLM generation via Ollama for zero-cost inference.
- Validated the retrieval approach with a controlled ablation: a hand-written hashing embedder scored false confidence up to 0.45 on nonsense queries, versus ~0.07–0.11 for real semantic embeddings.

charts.lk — Data Visualization Platform / *Client project at Cloud-eDesign — Next.js, MySQL, PayHere Payment Gateway*

- Built and help maintain charts.lk, a Next.js data-visualization platform for Sri Lanka backed by MySQL, including subscription billing via PayHere payment gateway integration.

Villa Management System / *Web Application — Spring Boot, MySQL, JSP, HTML, CSS, JavaScript* Mar 2024 – Jun 2024

- Built a full-stack web app for villa operations (bookings, service requests, staff coordination) using Spring Boot and MySQL for backend logic and data handling.

EDUCATION

BSc (Hons) in Information Technology, Specialising in Software Engineering / *Sri Lanka Institute of Information Technology (SLIIT)* Oct 2022 – Sep 2026

- Languages: English (Intermediate), Sinhala (Native)